

Prompt 1: How many pills are in this image? Answer with just a number.										Prompt 2: I need you to count every pill in this image carefully. Go through the image systematically, count each visible pill, then give me the final total.										Prompt 3: How many pills are in this image? After you give your answer, double-check by counting again and confirm or correct your number.										Prompt 3: How many pills are in this image? After you give your answer, double-check by counting again and confirm or correct your number.									
Image ID	True Count	Variable Tested	Model	Model Answer	Correct?	Off by	Model's reasoning notes			Image ID	True Count	Variable Tested	Model	Model Answer	Correct?	Off by	Model's reasoning notes			Image ID	True Count	Variable Tested	Model	Model Answer	Correct?	Off by	Model's reasoning notes			Image ID	True Count	Variable Tested	Model	Model Answer	Correct?	Off by	Model's reasoning notes		
1	7	baseline	ChatGPT	7	1	0				1	7	baseline	ChatGPT	8	0	1				1	7	baseline	ChatGPT							1	7	baseline	ChatGPT						
2	3	3 count	ChatGPT	3	1	0				2	3	3 count	ChatGPT	3	1	0				2	3	3 count	ChatGPT							2	3	3 count	ChatGPT						
3	5	5 count	ChatGPT	5	1	0				3	5	5 count	ChatGPT	5	1	0				3	5	5 count	ChatGPT							3	5	5 count	ChatGPT						
4	10	10 count	ChatGPT	10	1	0				4	10	10 count	ChatGPT	9	0	-1				4	10	10 count	ChatGPT							4	10	10 count	ChatGPT						
5	15	15 count	ChatGPT	12	0	-3				5	15	15 count	ChatGPT	14	0	-1				5	15	15 count	ChatGPT							5	15	15 count	ChatGPT						
6	7	spacing	ChatGPT	8	0	1				6	7	spacing	ChatGPT	7	1	0				6	7	spacing	ChatGPT							6	7	spacing	ChatGPT						
7	7	touching	ChatGPT	7	1	0				7	7	touching	ChatGPT	6	0	-1				7	7	touching	ChatGPT							7	7	touching	ChatGPT						
8	7	overlapping	ChatGPT	6	0	-1				8	7	overlapping	ChatGPT	6	0	-1				8	7	overlapping	ChatGPT							8	7	overlapping	ChatGPT						
9	7	background	ChatGPT	8	0	1				9	7	background	ChatGPT	7	1	0				9	7	background	ChatGPT							9	7	background	ChatGPT						
10	7	background	ChatGPT	7	1	0				10	7	background	ChatGPT	7	1	0				10	7	background	ChatGPT							10	7	background	ChatGPT						
11	7	lighting	ChatGPT	7	1	0				11	7	lighting	ChatGPT	7	1	0				11	7	lighting	ChatGPT							11	7	lighting	ChatGPT						
12	7	shadows	ChatGPT	7	1	0				12	7	shadows	ChatGPT	8	0	1				12	7	shadows	ChatGPT							12	7	shadows	ChatGPT						
13	7	color contrast	ChatGPT	8	0	1				13	7	color contrast	ChatGPT	8	0	1				13	7	color contrast	ChatGPT							13	7	color contrast	ChatGPT						
14	7	color uniformity	ChatGPT	8	0	1				14	7	color uniformity	ChatGPT	7	1	0				14	7	color uniformity	ChatGPT							14	7	color uniformity	ChatGPT						
15	7	normal	ChatGPT	7	1	0				15	7	normal	ChatGPT	7	1	0				15	7	normal	ChatGPT							15	7	normal	ChatGPT						
1	7	baseline	Claude	7	1	0				1	7	baseline	Claude	7	1	0				1	7	baseline	Claude							1	7	baseline	Claude						
2	3	3 count	Claude	3	1	0				2	3	3 count	Claude	3	1	0				2	3	3 count	Claude							2	3	3 count	Claude						
3	5	5 count	Claude	5	1	0				3	5	5 count	Claude	5	1	0				3	5	5 count	Claude							3	5	5 count	Claude						
4	10	10 count	Claude	10	1	0				4	10	10 count	Claude	10	1	0				4	10	10 count	Claude							4	10	10 count	Claude						
5	15	15 count	Claude	15	1	0				5	15	15 count	Claude	15	1	0				5	15	15 count	Claude							5	15	15 count	Claude						
6	7	spacing	Claude	6	0	-1				6	7	spacing	Claude	6	0	-1				6	7	spacing	Claude							6	7	spacing	Claude						
7	7	touching	Claude	6	0	-1				7	7	touching	Claude	4	0	-3				7	7	touching	Claude							7	7	touching	Claude						
8	7	overlapping	Claude	3	0	-4				8	7	overlapping	Claude	3	0	-4				8	7	overlapping	Claude							8	7	overlapping	Claude						
9	7	background	Claude	7	1	0				9	7	background	Claude	7	1	0				9	7	background	Claude							9	7	background	Claude						
10	7	background	Claude	7	1	0				10	7	background	Claude	7	1	0				10	7	background	Claude							10	7	background	Claude						
11	7	lighting	Claude	7	1	0				11	7	lighting	Claude	7	1	0				11	7	lighting	Claude							11	7	lighting	Claude						
12	7	shadows	Claude	7	1	0				12	7	shadows	Claude	7	1	0				12	7	shadows	Claude							12	7	shadows	Claude						
13	7	color contrast	Claude	7	1	0				13	7	color contrast	Claude	7	1	0				13	7	color contrast	Claude							13	7	color contrast	Claude						
14	7	color uniformity	Claude	7	1	0				14	7	color uniformity	Claude	7	1	0				14	7	color uniformity	Claude							14	7	color uniformity	Claude						
15	7	normal	Claude	6	0	-1				15	7	normal	Claude	6	0	-1				15	7	normal	Claude							15	7	normal	Claude						
1	7	baseline	Gemini	7	1	0				1	7	baseline	Gemini	7	1	0				1	7	baseline	Gemini							1	7	baseline	Gemini						
2	3	3 count	Gemini	3	1	0				2	3	3 count	Gemini	3	1	0				2	3	3 count	Gemini							2	3	3 count	Gemini						
3	5	5 count	Gemini	5	1	0				3	5	5 count	Gemini	5	1	0				3	5	5 count	Gemini							3	5	5 count	Gemini						
4	10	10 count	Gemini	10	1	0				4	10	10 count	Gemini	10	1	0				4	10	10 count	Gemini							4	10	10 count	Gemini						
5	15	15 count	Gemini	15	1	0				5	15	15 count	Gemini	15	1	0				5	15	15 count	Gemini							5	15	15 count	Gemini						
6	7	spacing	Gemini	7	1	0				6	7	spacing	Gemini	7	1	0				6	7	spacing	Gemini							6	7	spacing	Gemini						
7	7	touching	Gemini	7	1	0				7	7	touching	Gemini	7	0	0				7	7	touching	Gemini							7	7	touching	Gemini						
8	7	overlapping	Gemini	6	0	-1				8	7	overlapping	Gemini	7	0	0				8	7	overlapping	Gemini							8	7	overlapping	Gemini						
9	7	background	Gemini	7	1	0				9	7	background	Gemini	7	1	0				9	7	background	Gemini							9	7	background	Gemini						
10	7	background	Gemini	7	1	0				10	7	background	Gemini	7	1	0				10	7	background	Gemini							10	7	background	Gemini						
11	7	lighting	Gemini	7	1	0				11	7	lighting	Gemini	7	1	0				11	7	lighting	Gemini							11	7	lighting	Gemini						
12	7	shadows	Gemini	7	1	0				12	7	shadows	Gemini	7	1	0				12	7	shadows	Gemini							12	7	shadows	Gemini						
13	7	color contrast	Gemini	7	1	0				13	7	color contrast	Gemini	7	1	0				13	7	color contrast	Gemini							13	7	color contrast	Gemini						
14	7	color uniformity	Gemini	7	1	0				14	7	color uniformity	Gemini	7	1	0				14	7	color uniformity	Gemini							14	7	color uniformity	Gemini						
15	7	normal	Gemini	7	1	0				15	7	normal	Gemini	7	1	0				15	7	normal	Gemini							15	7	normal	Gemini						
Low counts are easy for all models																																							
High counts break some models (especially ChatGPT)																																							
SPACING / OVERLAP is the BIGGEST failure mode																																							
Background + lighting matter LESS than expected																																							
Color contrast matters (especially for ChatGPT)																																							
"Double-checking" is actually seeing better																																							
Even when models are explicitly asked to double-check their answer, they do not reliably correct mistakes. Instead, they often repeat the same incorrect reasoning in a more structured way																																							
Counting might improve when the task is decomposed into subgroups - colors, grains																																							
Overlap/touching = biggest failure																																							
High counts break some models (ChatGPT)																																							
Prompting does not reliably fix errors																																							
Models explain mistakes confidently instead of correcting them																																							
Scan the image from left to right and top to bottom. Count each pill as you encounter it and keep a running total. Then give the final count.																																							
Group the pills by color or type. Count how many are in each group, then add them together to get the total number of pills.																																							
First count the pills once.																																							
Then independently count them again in a different order.																																							
If the counts differ, explain why and give your best estimate.																																							
Biting																																							
dividing by region																																							
4																																							
9																																							